DRAGONN: Distributed Randomized Approximate Gradients of Neural Networks

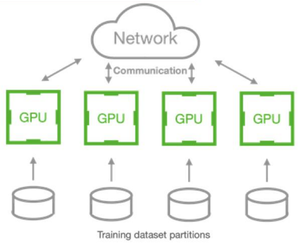
Zhuang Wang*, Zhaozhuo Xu*, Xinyu Crystal Wu, Anshumali Shrivastava, and T. S. Eugene Ng

* equal contribution

Introduction

Data-parallel Distributed Training (DDT)

Training with multiple GPUs



Communication bottleneck in DDT

Computation gets faster

- Advanced DNN accelerators
  - P100 -> V100 -> A100
- Advanced DNN compilers
  - XLA, TVM, etc.
- The single-GPU iteration time of ResNet50 has reduced by 22x

Network bandwidth can't catch up

- Slower-growing bandwidth
  - < 10x increase

Communication becomes the performance bottleneck

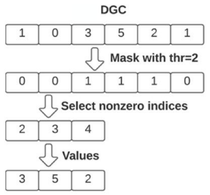
Gradient Sparsification (GS)

Select top-k gradients for synchronization

- Exact TopK GS
- Approximate TopK GS, e.g., DGC

Save up to 99.9% the gradient exchange

- Greatly reduce the communication time

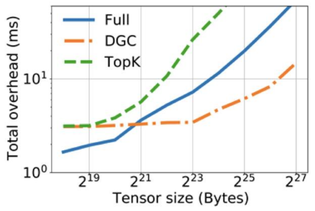


Method

Limitations of previous GS

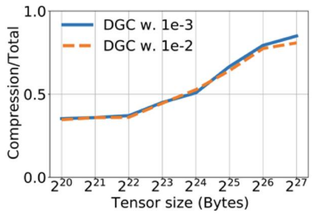
GS does not always help

- Full synchronization is better than DGC for smaller tensors



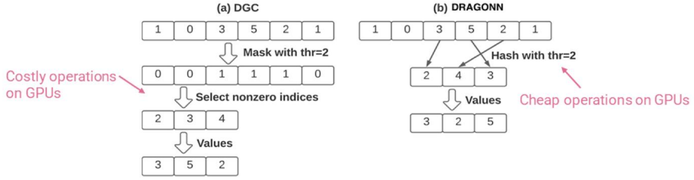
GS becomes the major bottleneck

- Compression time exceeds communication time



DRAGONN: a hashing-based compressor

Cheap encoding operations



Deploying DRAGONN in Practice

Efficiency-aware tensor selection for GS

- Apply DRAGONN to tensors only when it benefits the iteration time

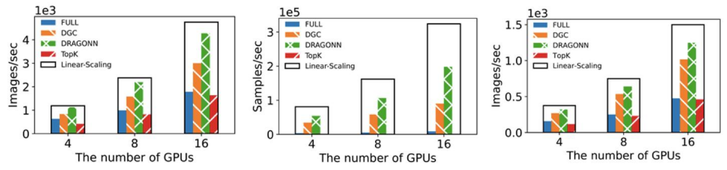
Sparse decoding

- Minimize the decoding overhead after communication

Experiment

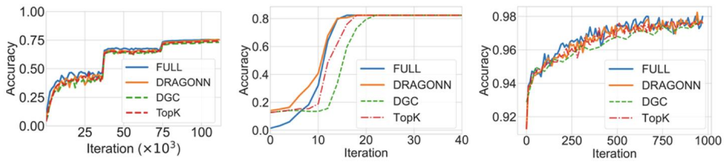
Evaluations

Training throughput



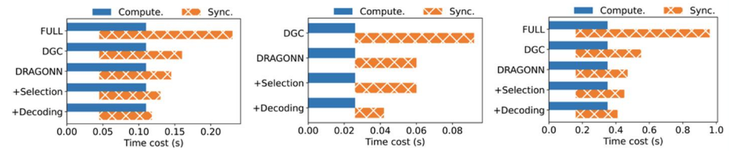
ResNet50 over ImageNet | XML over Wiki10-31K | ViT over ImageNet

Accuracy vs Iteration



ResNet50 | XML | ViT

Improvement breakdown



ResNet50 | XML | ViT

Acknowledgements

NSF, ONR DURIP, ONR BRC and Ken Kennedy Institute BP fellowship